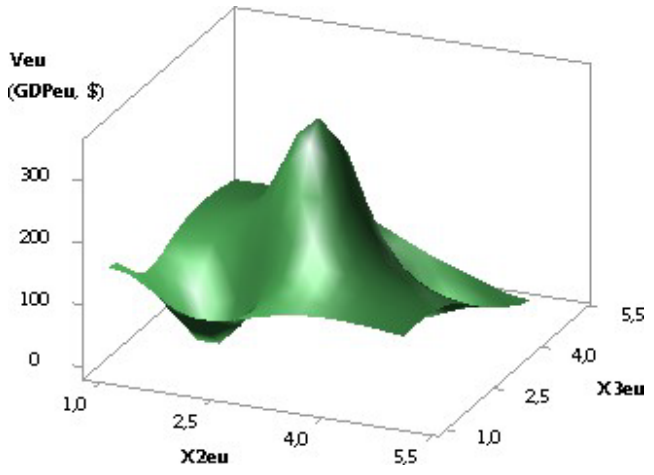
Э. А. ПИЛЬ

ТЕОРИЯ ЭКОНОМИЧЕСКИХ КРИЗИСОВ

III

THE THEORY OF ECONOMIC CRISES

III



САНКТ-ПЕТЕРБУРГ
2021

CONTENTS

INTRODUCTION 5

CHAPTER 1
CALCULATION OF UPPER CRITICAL VOLUME *Veu* AND X5*eu* VARIABLE VALUES OF ELLIPSOID ECONOMIC SHELL 6

1.1. Calculation of upper critical volume V_{eu} and $X5_{eu}$ variable values of ellipsoid economic shell 6
1.2. Summary tables for upper critical volume V_{eu} and $X5_{eu}$ variable values of ellipsoid economic shell 58
1.3. Calculation of upper critical volume V_{eu} and $X5_{eu}$ variable values given considerable decreases in values of all variables 66
1.4. Calculation of upper critical volume V_{eu} and $X5_{eu}$ variable values given considerable increases in values of all variables 67
1.5. Calculation of upper critical volume V_{eu} and $X5_{eu}$ variable values given increased number of decimals in variable $X4_{eu}$ values 69
1.6. Calculation of upper critical volume V_{eu} and $X5_{eu}$ variable values given considerable decreases in values of all variables and increased number of decimals in $X4_{eu}$ variable values 70
1.7. Calculation of upper critical volume V_{eu} and $X5_{eu}$ variable values given considerable increases in values of all variables and increased number of decimals in $X4_{eu}$ variable values 72
1.8. Calculation of upper critical value V_{etu1} of theoretically attainable unit volume and $X5_{eu}$ variable values of ellipsoid economic shell 73

CONCLUSIONS 75

CHAPTER 2
CALCULATION OF UPPER CRITICAL VOLUME *Veu* AND X6*eu* VARIABLE VALUES OF ELLIPSOID ECONOMIC SHELL 76

2.1. Calculation of upper critical volume V_{eu} and $X6_{eu}$ variable values of ellipsoid economic shell 76
2.2. Summary tables for upper critical volume V_{eu} and $X6_{eu}$ variable values of ellipsoid economic shell 128
2.3. Calculation of upper critical volume V_{eu} and $X6_{eu}$ variable values given considerable increases in values of all variables 135
2.4. Calculation of upper critical volume V_{eu} and $X6_{eu}$ variable values given increased number of decimals in variable $X4_{eu}$ values 137
2.5. Calculation of upper critical volume V_{eu} and $X6_{eu}$ variable values given considerable decreases in values of all variables and increased number of decimals in variable $X4_{eu}$ values 139
2.6. Calculation of upper critical volume V_{eu} and $X6_{eu}$ variable values given considerable decreases in values of all variables and increased number of decimals in variable $X4_{eu}$ values 139
2.7. Calculation of upper critical volume V_{eu} and $X6_{eu}$ variable values given considerable increased values of all variables and increased number of decimals in variable $X4_{eu}$ values 141
2.8. Calculation of upper critical value V_{etu1} of theoretically attainable unit volume and variable $X6_{eu}$ values of ellipsoid economic shell 142

CONCLUSIONS 144

CHAPTER 3
CALCULATION OF LOWER CRITICAL VOLUME *Vel* AND X4*el* VARIABLE VALUES OF ELLIPSOID ECONOMIC SHELL 145

3.1. Calculation of lower critical volume V_{el} and $X4_{el}$ variable values of ellipsoid economic shell 145
3.2. Summary tables for lower critical volume V_{el} and $X4_{el}$ variable values of ellipsoid economic shell 197
3.3. Calculation of lower critical volume V_{el} and $X4_{el}$ variable values given considerable decreases in values of all variables 203
3.4. Calculation of lower critical volume V_{el} and $X4_{el}$ variable values given considerable increases in values of all variables 205
3.5. Calculation of lower critical value V_{etl1} of theoretically attainable unit volume and $X4_{el}$ variable values of ellipsoid economic shell 206

CONCLUSIONS 208

CHAPTER 4
CALCULATION OF LOWER CRITICAL VOLUME *Vel* AND X5*el* VARIABLE VALUES OF ELLIPSOID ECONOMIC SHELL 209
4.1. Calculation of lower critical volume V_{el} and $X5_{el}$ variable values of ellipsoid economic shell 209
4.2. Summary tables for lower critical volume V_{el} and $X5_{el}$ variable values of ellipsoid economic shell 258
4.3. Calculation of lower critical volume V_{el} and $X5_{el}$ variable values given considerable decreases in values of all variables 264
4.4. Calculation of lower critical volume V_{el} and $X5_{el}$ variable values given considerable increases in values of all variables 265
4.5. Calculation of lower critical value V_{etl1} of theoretically attainable unit volume and $X5_{el}$ variable values of ellipsoid economic shell 267
CONCLUSIONS 269

CHAPTER 5
CALCULATION OF VOLUME Δ*Veul* DOMAIN OF ELLIPSOID ECONOMIC SHELL AND X4*eul* VARIABLE VALUES 270
5.1. Calculation of lower critical volume ΔV_{eul} domain of ellipsoid economic shell and $X4_{eul}$ variable values 270
5.2. Summary tables for volume ΔV_{eul} domain of ellipsoid economic shell and $X4_{eul}$ variable values 320
CONCLUSIONS 327

CHAPTER 6
CALCULATION OF VOLUME Δ*Veul* DOMAIN OF ELLIPSOID ECONOMIC SHELL AND X5*eul* VARIABLE VALUES 328
6.1. Calculation of lower critical volume ΔV_{eul} domain of ellipsoid economic shell and $X5_{eul}$ variable values 328
6.2. Summary tables for volume ΔV_{eul} domain of ellipsoid economic shell and $X5_{eul}$ variable values 377
CONCLUSIONS 384
LIST OF REFERENCES 385
BIOGRAPHY 400

'For me a word of wisdom is more valuable than any gold.'
Democritus

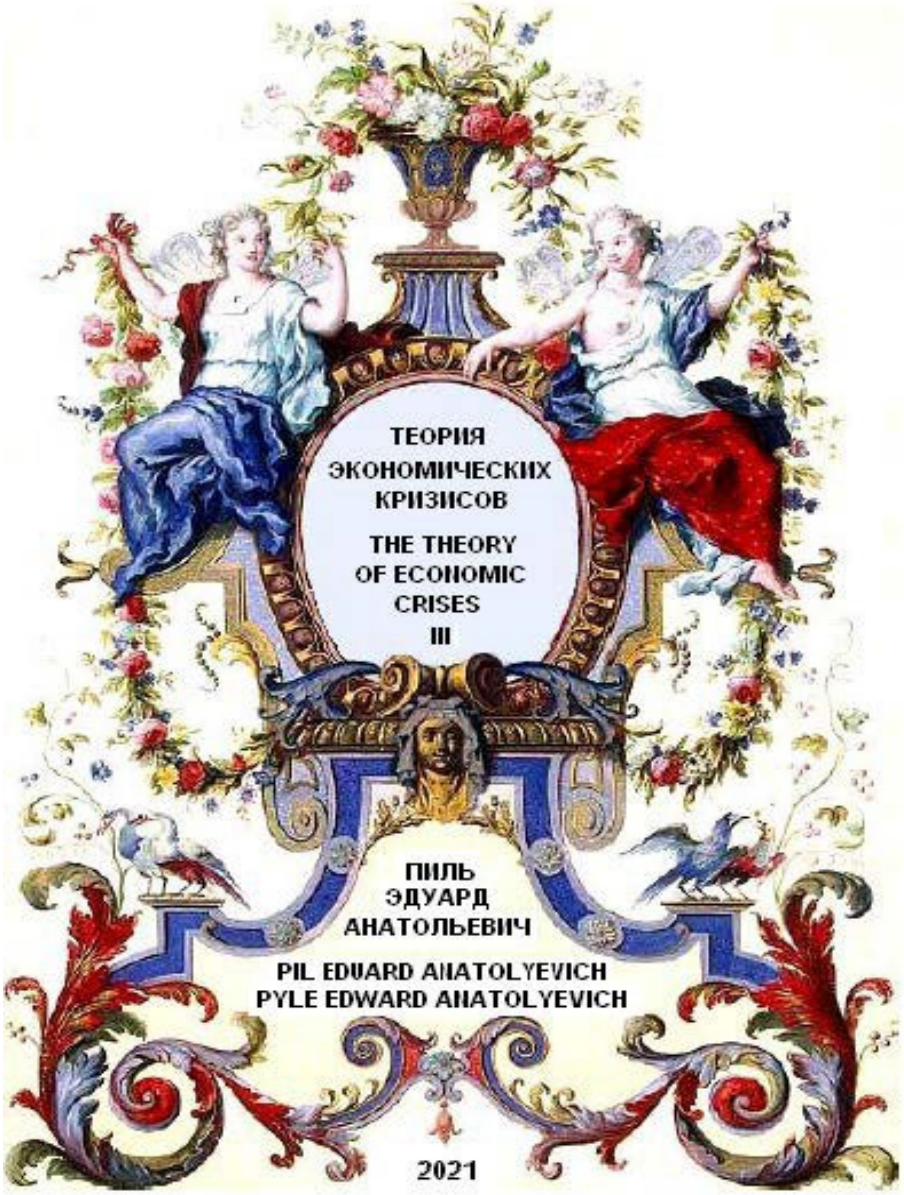
INTRODUCTION

The issue of economic and financial crises has been attended in a wide range of studies offering various solutions, none, however, allowing for a complex approach. Thus, the present study offers to apply the theory of shells to determine and describe the processes in crisis economy. The theory, so applied, allows for predicting theoretically possible scenarios, performing calculations and, based on these, developing ways to resolve crisis conditions in an economy. Governments may use the instruments offered and select a most suitable and effective solution either maintaining the current level or even increasing the country's GDP.

The book is dedicated to my son *Mikhail* and grandson *Stepan*, and to my late beloved parents and elder brother.

Wishing you all the best.

Yours sincerely, Author



[127]